Sohan Kathari

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Education

Carnegie Mellon University Pittsburgh, PA
Bachelor of Science in Computer Science, GPA:4.0/4.0 Expected May 2030
Coursework: Data Structures & Algorithms, Discrete Mathematics, Linear Algebra, Calculus

Skills

Languages: Python, C++, C, Java, TypeScript, JavaScript
Technologies: React, Next.js, React Native, Flask, Firebase, Git, Linux, Vercel, Expo
AI/ML: PyTorch, pandas, NumPy, scikit-learn, Whisper, OpenAI APIs, MediaPipe, CNNs, RNNs, Codex, Claude Code

Honors

Valedictorian (GPA 4.0, Rank 1/872), HackCMU 2026 2nd Place, Texas Congressional Debate Champion

Experience

Carnegie Mellon Racing Club – *Driverless Software Developer* Sep 2026 – Present
- Contributing to a **40+** contributor autonomous racing codebase, developing trajectory-generation and map-editing tools for configurable path geometry and manual route optimization
- Designing virtual-cone generation for skidpad planning by tracing localization, perception, and controller data flows to provide consistent track boundaries where physical cones are absent
- Implementing C1-continuous path transitions with tangent-aligned Bézier curves and debugging visualization issues identified through vehicle track testing

SpeakSmart – *Founding Developer* Jan 2025 – Jun 2026
- Engineered and deployed a full-stack speech-analysis platform with Next.js and TypeScript, integrating transcription, LLM analysis, and computer vision into an end-to-end video-processing pipeline
- Optimized asynchronous video processing and selective frame sampling to analyze **100+ recordings** in **<60 seconds** per video
- Deployed on Vercel and scaled the platform to **7,200+ visits**, iterating on the product through community workshops and user feedback

AllergyShield – *Founding Developer* Aug 2025 – Apr 2026
- Created and shipped an iOS food-allergen detection app with React Native, TypeScript, Expo, Firebase, and multimodal LLM APIs to analyze multilingual food labels and generate personalized allergen risk assessments
- Architected Firebase Auth, Firestore, and Cloud Storage infrastructure for authentication, allergen profiles, image uploads, and persistent scan history
- Released on the Apple App Store, reaching **150+ users** and piloted across **5 retail locations**

Leadership & Activities

Flower Mound Speech and Debate – *Competitor and Mentor* Aug 2023 – May 2026
- Won the TFA Texas State Championship in Congressional Debate; won **9 tournaments** and reached **42 finals** across Congressional Debate and Extemporaneous Speaking; placed Top 18 in Domestic Extemp at TFA State twice
- Researched complex policy and current-events topics, synthesized evidence into persuasive arguments, and delivered speeches under strict preparation and time constraints
- Mentored **40+ students** on argumentation, research, and delivery, supporting a program with **55 state qualifications**

Projects

GlucoPilot – 2nd Place, HackCMU Hackathon Sep 2026
- Built a personalized glucose-forecasting system integrating CGM, insulin, meal, exercise, sleep, caffeine, and alcohol data using ODE simulation and UKF state estimation
- Validated predictions across **30 virtual simulated patients** and integrated multimodal sensor, image, and text inputs for meal and activity modeling

Solar Energy Anomaly Detection – *Rayfield Systems / HUVTSP* Jun 2025 – Aug 2025
- Developed a Python and Flask pipeline to ingest solar-generation data, detect abnormal production patterns using Isolation Forest, and surface anomalies for manual review
- Processed and analyzed real-world operational energy data using pandas, achieving **95% precision** against manually labeled anomalies

AuthenAI Deepfake Detection System Dec 2024 – Jun 2025
- Built a CNN + RNN multimodal Deepfake detection system for video, audio, and hybrid inputs
- Achieved a **0.90+ average F1 score** across FaceForensics++, DeepFakeTIMIT, and Celeb-DF datasets